

WASP-104b

R-2273

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_221217 · created 2026-08-16T23:08:37+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459930.967 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.117 +/- 0.011
Transit depth (Rp/Rs)^2	1.38 +/- 0.25 [%]
Orbital Inclination (inc)	85.22 +/- 1.33
Airmass coefficient 1 (a1)	1.015 +/- 0.01
Airmass coefficient 2 (a2)	-0.012 +/- 0.011
Scatter in the residuals of the lightcurve fit is	1.02 %
Best Comparison Star	#2 - [374, 163]
Optimal Aperture	3.47
Optimal Annulus	14.34
Transit Duration (day)	0.0831 +/- 0.008

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.117 ± 0.011 vs 0.1214 (deviation 0.39σ)
Duration fit vs published	0.0831 d vs 0.0742 d (deviation 1.08σ)
Mid-time O–C	0.6 min
Depth SNR	10.3
Residual scatter	1.02 %

Light curve

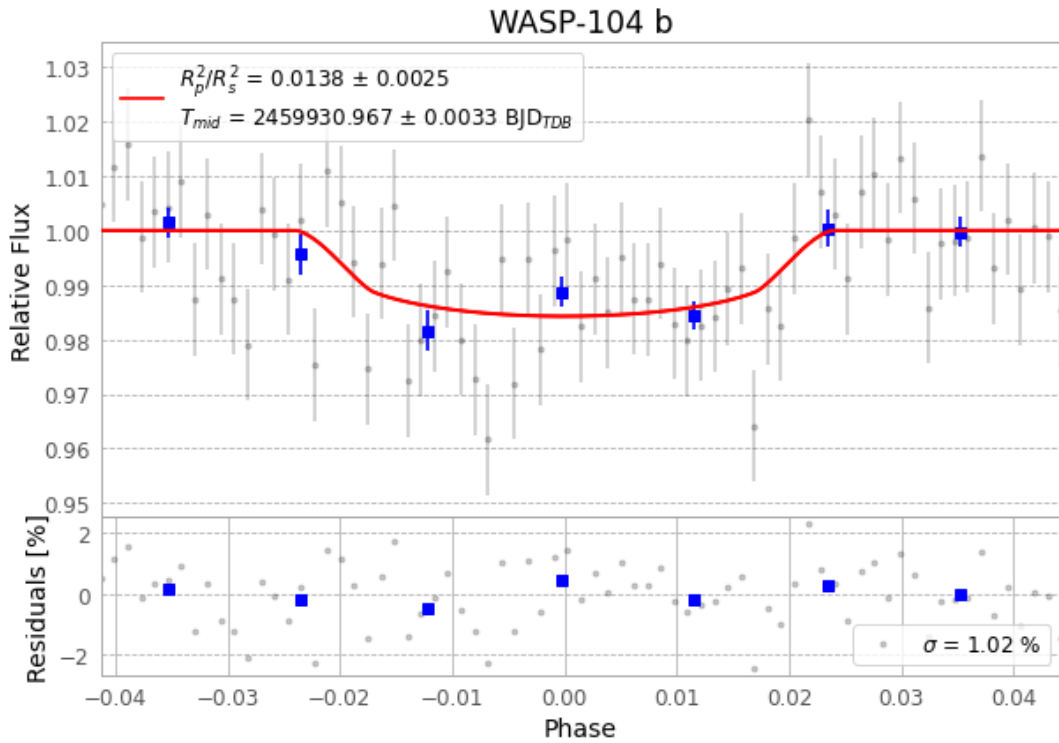


Figure 1 — FinalLightCurve_WASP-104 b_2022-12-17.png (SHA-256 858404854e3e17a8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [360, 121], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 510f9e91cd2b9e3c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

73 FITS frames (48 MB), WASP-104221217092415.FITS → WASP-104221217130015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-221217.log (629809979961...), FinalLightCurve_WASP-104 b_2022-12-17.png (858404854e3e...), FinalLightCurve_WASP-104 b_2022-12-17.csv (3c66af9554cf...)

Compute resources

Wall time 127.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 23:08Z.